

TABLE II
Test Results

IEC name	LR6 (Alkaline)			R6 (Zinc-carbon)				FR6 (Li-FeS ₂)	
	Duracell Copper Top	Sony Stamina Plus	Eveready Ultima	Panasonic	Nippo Gold	Eveready	Nippo	Energizer Ultimate Lithium	Duracell Ultra Lithium
Model number	MN1500	AM3	2115	UM-3NG	UM-3DG	1015	UM-3U	L91	LF1500
Price (₹)	28	20	20	10	10	11	7	175	170
Sustained duration (in hours)	3.8258	3.4108	3.8944	0.8736	0.8533	0.7144	0.4253	7.24	8
Load	2.5Ω								
Minimum predefined voltage	0.8V								

hydrogen-absorbing alloy instead of cadmium, being in essence a practical application of nickel-hydrogen battery chemistry. An NiMH cell can have two to three times the capacity of an equivalent-sized NiCd, and its energy density approaches that of a lithium-ion cell.

Lithium ion. A lithium-ion cell (sometimes called Li-ion battery) is a rechargeable battery cell in which lithium ions move from the negative electrodes to the positive electrodes during discharge and back when charging. These batteries use an intercalated lithium compound as the electrode material, compared to the metallic lithium used in non-rechargeable lithium batteries.

Lead-acid. These are the most popular rechargeable battery cells worldwide. Both the battery product and the manufacturing process are proven, economical and reliable. However, as these batteries are heavy and use a liquid electrolyte, they are not used in portable consumer electronics.

Technical standards for battery sizes and types are published by standards organisations such as International Electrotechnical Commission (IEC) and American National Standards Institute (ANSI). Some sizes of batteries are D, C, AA, AAA, AAAA, A23, 9V and CR2032.

Accelerated testing

Different AA-size primary battery cells were chosen for testing in EFY lab. These cells come with a variety of chemistries and therefore the difference in their prices. You can identify

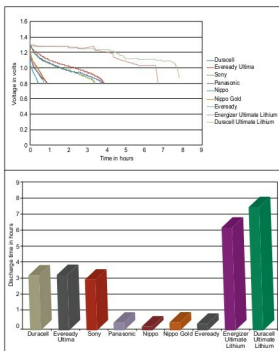


Fig. 4: Real-time data (up to 0.8V)

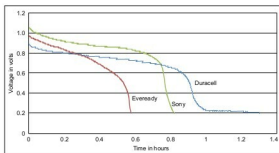


Fig. 5: Real-time data (0.8V to 0.2V)

different cells with their IEC names as shown in Table I.

Fig. 2 shows the test setup for accelerated testing of various AA-size primary batteries to have a rough idea about their performance vis a vis their price. Our testing was simple. Different batteries were put on

the same load (2.5 ohms). The voltage of each battery was continuously fetched by a microcontroller (PIC16F877A). The microcontroller sent the real-time voltage levels of all the battery cells to a MATLAB program running on a computer via serial port. The program saved these values in an Excel sheet for all the cells. The program stopped the process when all the batteries' voltage fell below the minimum predefined level. The program provided the result in number of hours it took to discharge up to that level. Here 0.8V was taken as the minimum predefined value.

Table II shows the test results for different makes and chemistries. The data shows that Eveready Ultima (LR6) gave the best performance per unit price as shown in Fig. 3.

Fig. 4 shows the real-time graphs generated by the software. But when the test was run further in the range of 0.8V to 0.2V, the Duracell sustained much longer amongst all alkaline (LR6) batteries. Other batteries degraded very quickly after 0.8V but Duracell sustained much longer as shown in Fig. 5. ●

The author is a B.Tech in electronics and communications from SRCEM College, Gwalior